

CANDIDATE NAME: _____

INDEX NUMBER _____



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2018

H2 BIOLOGY
Paper 1 Multiple Choice

CG _____

9744/01

Friday
21 September 2018

Additional Materials: Multiple Choice Answer Sheet

1 hour

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CG in the spaces at the top of this page.

On the Multiple Choice Answer Sheet, write your name, subject title, test name and CG. For your index number, write your full NRIC number. Shade the corresponding lozenges on the Answer Sheet according to the instructions given by the invigilators.

There are **thirty questions** on this paper. Answer **all** questions. For each question, there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

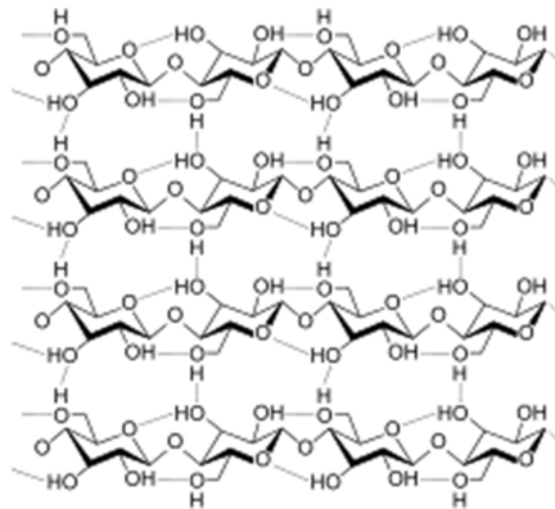
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

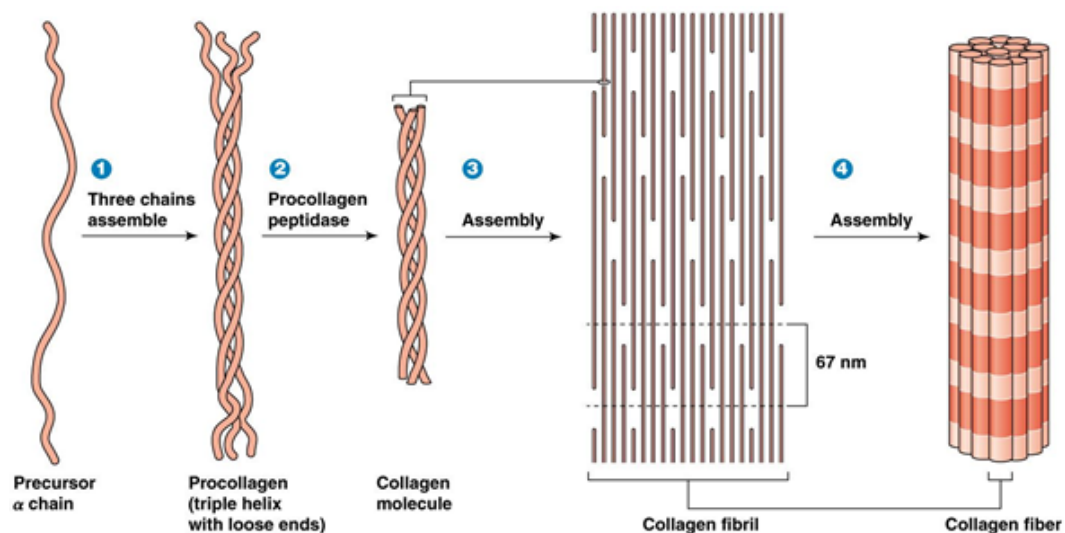
AT THE END OF THE EXAMINATION, SUBMIT BOTH THE MULTIPLE CHOICE ANSWER SHEET AND THE QUESTION PAPER.

1. The following biomolecule is



- A not a protein because of the presence of phenyl groups.
- B a protein because of the presence of peptide linkages.
- C not a carbohydrate because of the presence of regular repeated folding.
- D a carbohydrate because of the presence of glycosidic linkages.

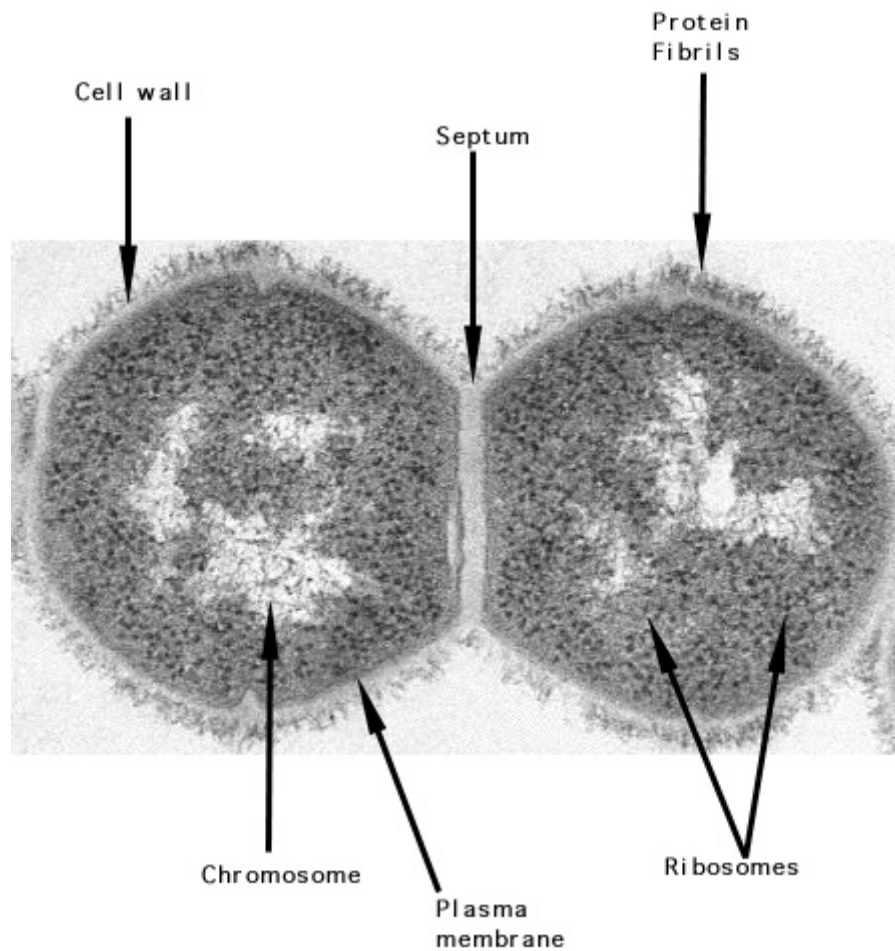
2. The following is a diagrammatic representation of the macromolecular assembly of a collagen fibre.



Which of the following descriptions is incorrect?

- A Every third amino acid is glycine as it has the smallest R group.
- B Collagen fibrils can be held together by covalent linkages.
- C Procollagen is a secondary protein due to regular coiling of three polypeptides.
- D Many procollagen molecules associate to create collagen.

3. The following is an electron micrograph of a living cell that is dividing.



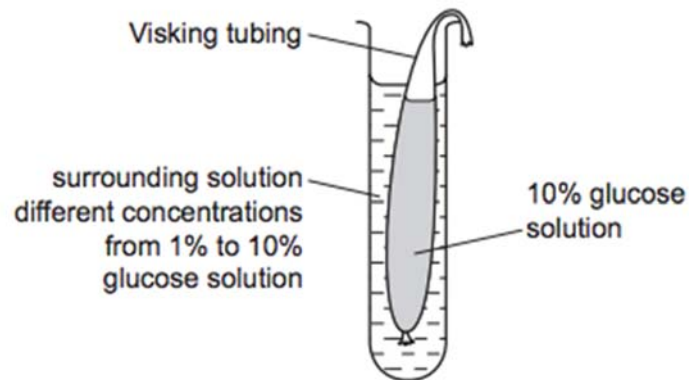
The following are some possible statements about the daughter cells:

- 1 These are a pair of plant cells as the septum represents the cell plate of the dividing cell.
- 2 These are a pair of bacterial cells as they have chromosomes and ribosomes.
- 3 These cannot be bacterial cells as they have a cell wall.
- 4 These are a pair of animal cells as they have proteins on their outer surface.
- 5 These cannot be animal cells as the ribosomes are located alongside the chromosomes.
- 6 These cannot be bacterial cells as circular DNA is not evident.

Which of the statements about these cells are correct?

- A** 1 and 3
- B** 1 and 6
- C** 3 and 4
- D** 2 and 5

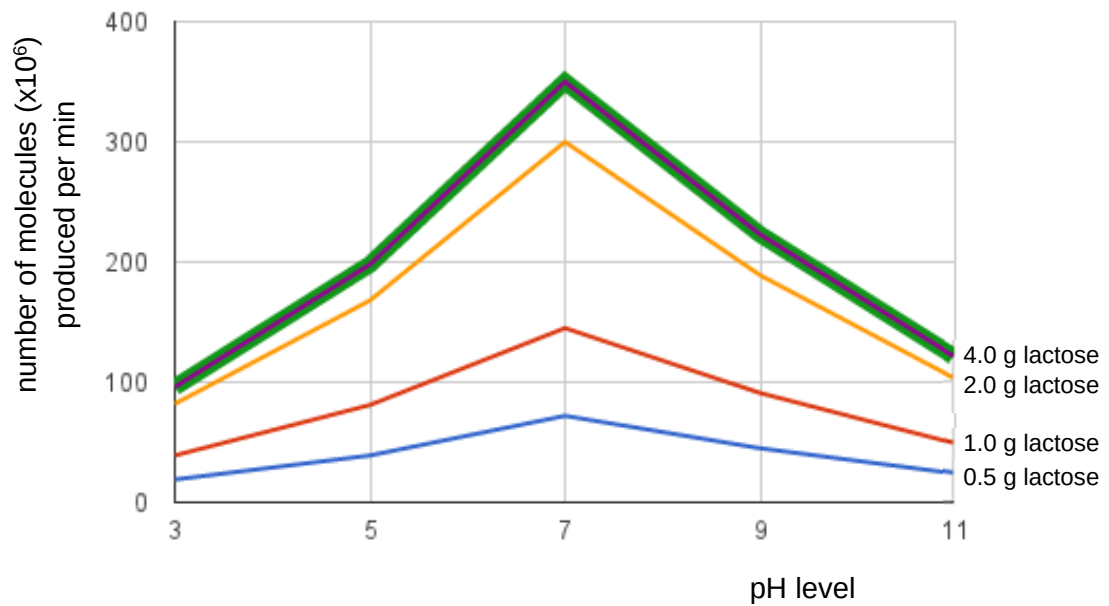
4. The diagram shows apparatus set up to investigate the effect of changing the concentration of glucose in the surrounding solution on the movement of molecules through a selectively permeable membrane (Visking tubing) in 15 minutes.



As the concentration of glucose solution in the surrounding solution increases, which statements are correct?

- 1 Net diffusion of water increases.
 - 2 Glucose molecules reach an equilibrium quicker.
 - 3 There is less change in the volume of surrounding solution.
 - 4 Net diffusion of glucose increases.
- A** 1 and 3 only
B 2 and 3 only
C 1, 2 and 4 only
D 1, 2, 3 and 4

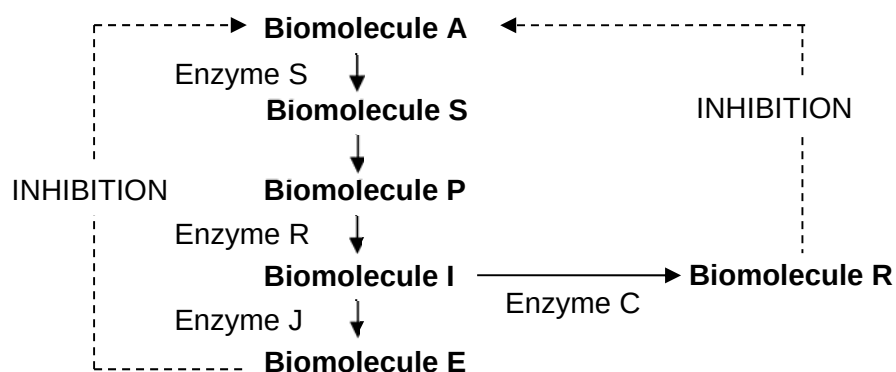
5. The following is a graph showing the number of molecules of product produced from the digestion of lactose at different pH levels.



Which of the following is **not** a valid conclusion from this graph?

- A Enzyme molecules are fully denatured at extreme pH levels.
- B The reaction rate is likely to level off past a certain substrate concentration.
- C pH level of 7 is the optimal pH for this enzyme.
- D Increasing substrate concentration increases the rate of enzyme reaction.

6. A hypothetical metabolic pathway is shown below.



Which changes in enzyme activity will result in the greatest increase in the yield of Biomolecule R?

	Enzyme	Change in activity	Enzyme	Change in activity
A	S	Decrease	J	Decrease
B	R	Increase	J	Decrease
C	J	Increase	C	Increase
D	C	Decrease	R	Decrease

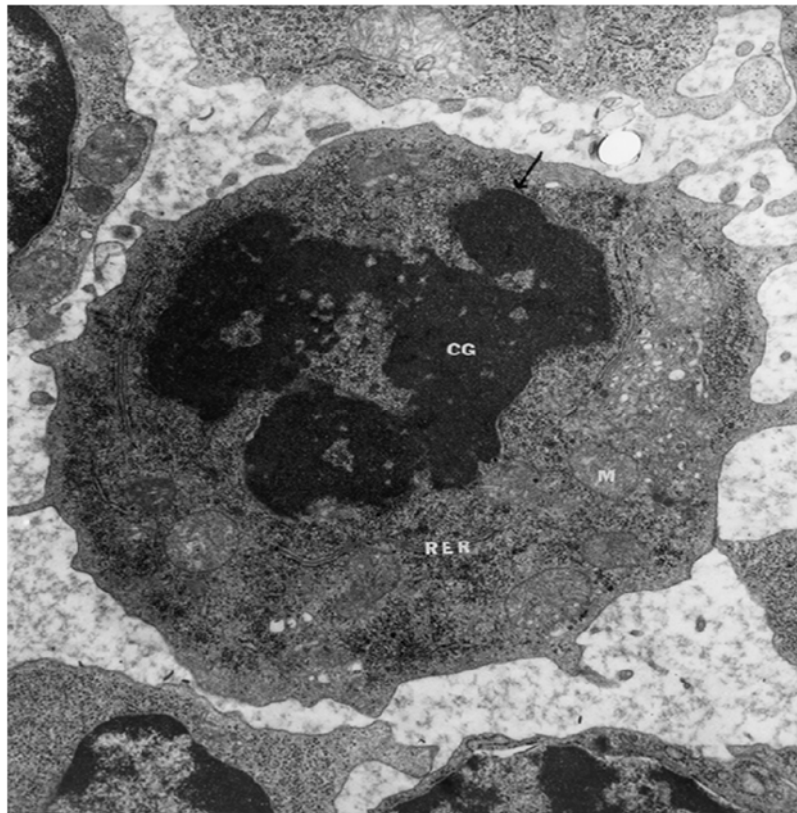
7. The following are some statements related to cell and nuclear division.

- Both haploid and diploid cells have homologous chromosomes.
- Fertilisation doubles the chromosome number while subsequent mitotic division temporarily doubles it further.
- Meiotic division halves amount of DNA twice while mitotic division halves the amount of DNA only once.
- The cell cycle of a specific cell includes either meiotic or mitotic division but never both at the same time.

Which of the above statements is/are **incorrect**?

- 1 only
- 1 and 2 only
- 2 and 3 only
- 2, 3 and 4 only

8. The following depicted cell is undergoing cell division.

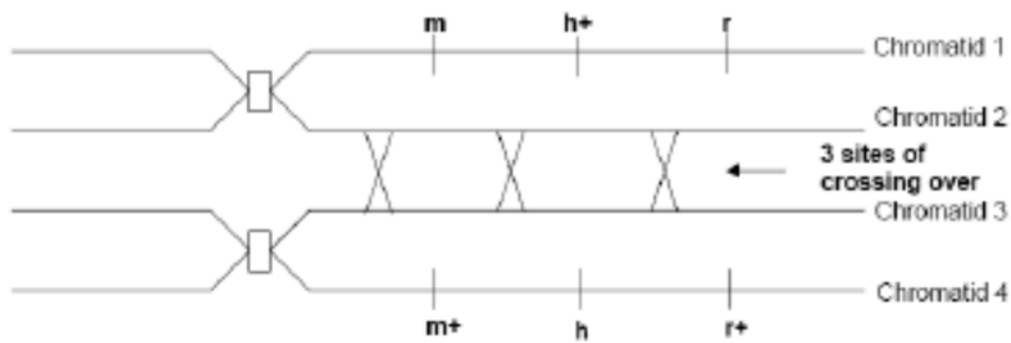


(CG = chromatin granules, RER = rough endoplasmic reticulum, M = mitochondrion)

Based on this electron micrograph, which of the following statements about this cell is accurate?

- A** The cell is undergoing meiosis 1 as the chromatin is separating into four components.
- B** The cell is undergoing prophase of meiosis as chromosomes are synapsing.
- C** The cell is undergoing interphase as the chromosomes are replicating.
- D** The cell is undergoing prophase as the nuclear envelope has broken down.

9. The diagram below shows a pair of homologous chromosomes during prophase of meiosis.



Determine the resulting segregation of the named alleles.

- | | |
|--|--|
| <p>A chromatid 1: $m\ h^+\ r$
 chromatid 2: $m\ h\ r$
 chromatid 3: $m^+\ h^+\ r^+$
 chromatid 4: $m^+\ h\ r^+$</p> | <p>C chromatid 1: $m\ h^+\ r$
 chromatid 2: $m^+\ h^+\ r^+$
 chromatid 3: $m\ h\ r$
 chromatid 4: $m^+\ h\ r^+$</p> |
| <p>B chromatid 1: $m\ h^+\ r$
 chromatid 2: $m^+\ h\ r^+$
 chromatid 3: $m\ h^+\ r$
 chromatid 4: $m^+\ h\ r^+$</p> | <p>D chromatid 1: $m\ h^+\ r$
 chromatid 2: $m^+\ h^+\ r$
 chromatid 3: $m\ h\ r^+$
 chromatid 4: $m^+\ h\ r^+$</p> |

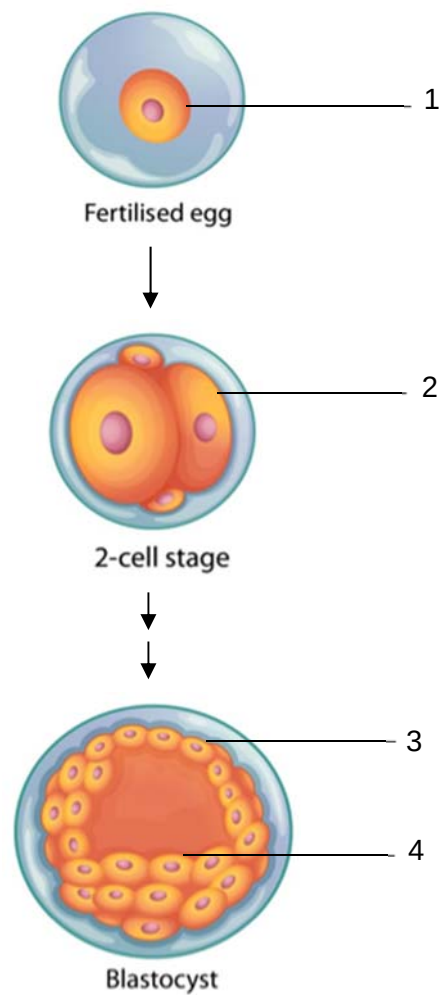
10. Scientists have made a nucleic acid (HNA) that has a sugar with the same number of carbon atoms as glucose instead of deoxyribose. Although genetic information can be stored by HNA, naturally occurring DNA polymerase cannot replicate HNA.

Which statements could explain why naturally occurring DNA polymerase cannot replicate HNA?

- 1 DNA polymerase cannot form bonds between the sugars of two HNA nucleotides.
- 2 DNA polymerase cannot form hydrogen bonds between two HNA nucleotides.
- 3 HNA nucleotides do not fit into the active site of DNA polymerase.
- 4 The shape of an HNA nucleotide is slightly larger than that of a DNA nucleotide.

- A** 1, 2, 3 and 4
B 1 and 4 only
C 2 and 3 only
D 3 and 4 only

11. The figure below shows some stages of embryonic development. Four different cells are labelled 1, 2, 3 and 4 as shown.



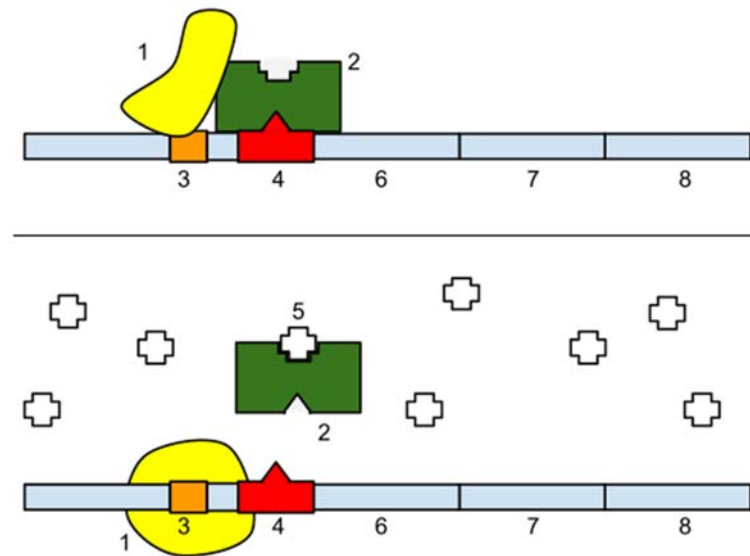
Which of the following statements is incorrect?

- A Cells 1 and 2 are totipotent.
- B Cells 3 and 4 are pluripotent.
- C Cell 3 is a result of mitoses involving Cell 1.
- D Cells 2 and 4 can give rise to adult stem cells.

12. Which of the following is an example of translational control of gene expression?

- A Activation of proteins by folding or enzymatic cleavage
- B Addition of chemical groups such as phosphate groups to free amino acids in the cytoplasm
- C Binding of protein factors to specific sequences in mRNA preventing ribosomes from attaching.
- D Formation of disulfide bridges in the protein being formed

13. A simplified figure of the lac operon is shown below. Several structures have been numbered 1 to 8.



Which of the following correctly identifies some of the structures above?

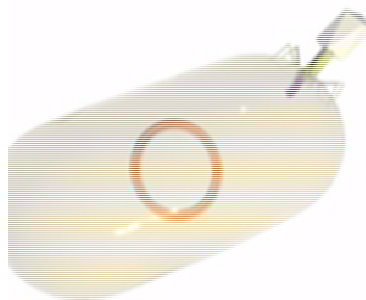
- A 1 is the lac repressor, 3 is the operator and 5 is allolactose.
- B 2 is an enzyme, 4 is the operator and 7 is a structural gene.
- C 2 is a catabolite activator protein, 5 is cyclic AMP and 8 is a structural gene.
- D 3 is the promoter, 5 is a carbohydrate and 6 is a structural gene.

14. Which of the following pairs of statements are correct of generalised and specialised transduction?

	Generalised	Specialised
1	Transfers any bacterial DNA	Transfers a specific set of bacterial genes
2	Contains a hybrid chromosome in its capsid	Contains only bacterial chromosome in its capsid
3	Host cell will die	Host cell may die
4	Viral genome is not transcribed	Viral genome is transcribed
5	Only involves lytic cycle	Only involves lysogenic cycle
6	Viral DNA is replicated by host cell machinery	Viral DNA is replicated by binary fission

- A 1, 3 and 6 only
- B 1, 4 and 5 only
- C 2, 3 and 6 only
- D 3, 4 and 5 only

15. The following shows a virus infecting a bacterial cell.



After the step shown above, the viral genome could possibly

- A be reverse transcribed.
- B replace the host genome.
- C be packaged into a new capsid coat.
- D be integrated into the host bacterial genome.

16. SR scientists carried out an investigation into the effects of increasing temperature from 37 °C to 55 °C on the rates of glycolysis and Krebs cycle in animal cells.

They found that the rate of glycolysis decreased by 10% upon the increase in temperature while the rate of Krebs cycle decreased by 6%.

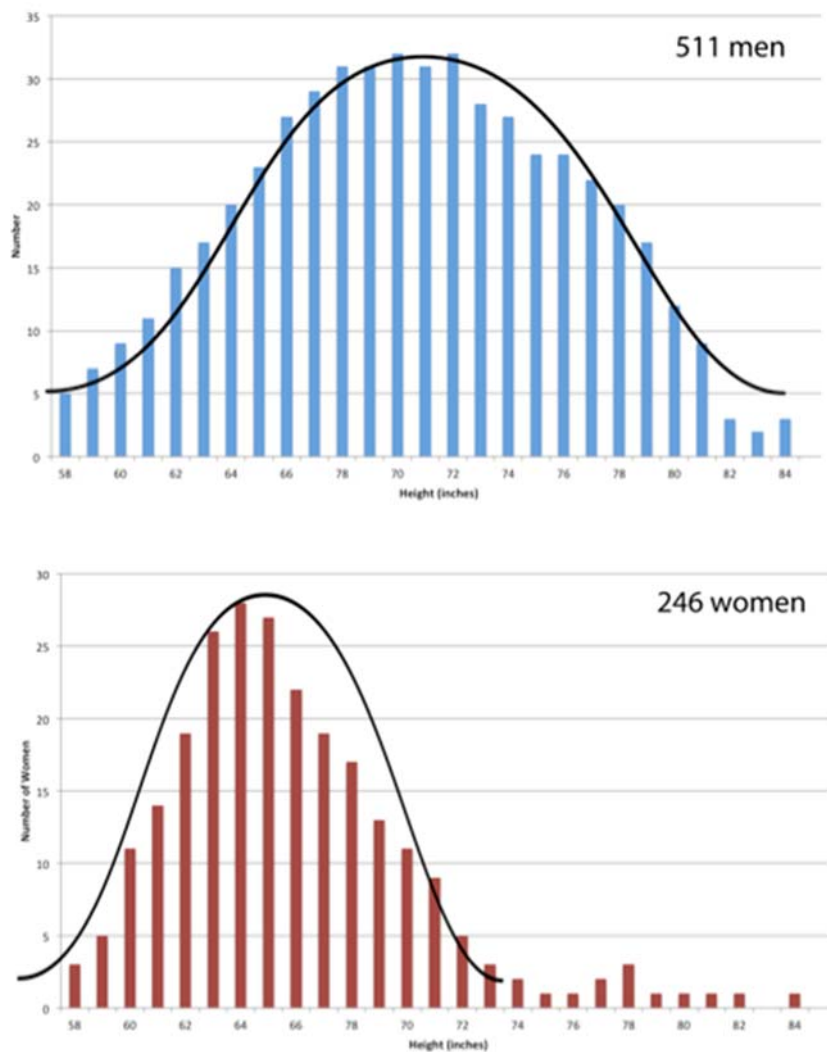
A Student's t-test was conducted, and the calculated t value at degree of freedom = 8 was 2.392.

v	α													
	0.40	0.30	0.20	0.15	0.10	0.05	0.025	0.02	0.015	0.01	0.0075	0.005	0.0025	0.0005
1	0.325	0.727	1.376	1.963	3.078	6.314	12.706	15.895	21.205	31.821	42.434	63.657	127.322	636.590
2	0.289	0.617	1.061	1.386	1.886	2.920	4.303	4.849	5.643	6.965	8.073	9.925	14.089	31.598
3	0.277	0.584	0.978	1.250	1.638	2.353	3.182	3.482	3.896	4.541	5.047	5.841	7.453	12.924
4	0.271	0.569	0.941	1.190	1.533	2.132	2.776	2.999	3.298	3.747	4.088	4.604	5.598	8.610
5	0.267	0.559	0.920	1.156	1.476	2.015	2.571	2.757	3.003	3.365	3.634	4.032	4.773	6.869
6	0.265	0.553	0.906	1.134	1.440	1.943	2.447	2.612	2.829	3.143	3.372	3.707	4.317	5.959
7	0.263	0.549	0.896	1.119	1.415	1.895	2.365	2.517	2.715	2.998	3.203	3.499	4.029	5.408
8	0.262	0.546	0.889	1.108	1.397	1.860	2.306	2.449	2.634	2.896	3.085	3.355	3.833	5.041
9	0.261	0.543	0.883	1.100	1.383	1.833	2.262	2.398	2.574	2.821	2.998	3.250	3.690	4.781
10	0.260	0.542	0.879	1.093	1.372	1.812	2.228	2.359	2.527	2.764	2.932	3.169	3.581	4.587

Using the table above, which statement below gives an appropriate conclusion at the 5% significance level?

- A An increase in temperature to 55 °C has a greater effect on glycolysis.
- B An increase in temperature to 55 °C has a greater effect on the Krebs cycle.
- C Glycolysis and Krebs cycle are affected similarly by the increase in temperature to 55 °C.
- D Glycolysis and Krebs cycle are not significantly affected by the increase in temperature to 55 °C.

17. The following graph depicts the distribution of human heights in a sample of 20-year old males and females in the US.



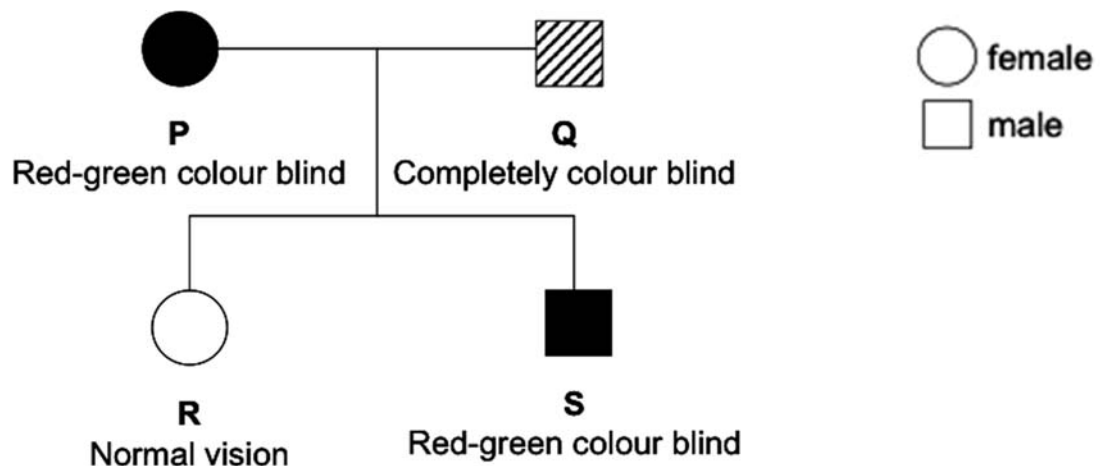
Based on the information above and your own knowledge, which of the following is **not** a valid inference?

- A Human height is controlled by more than one gene locus.
- B Human height involves multiple alleles or epistasis.
- C Both the environment and genes have an effect on human height.
- D Women have a lower mean height than men.

18. Red-green colour blindness is controlled by a gene on the X chromosome. The allele for colour blindness, **g**, is recessive to the allele for normal colour vision, **G**.

Complete colour blindness is controlled by a different gene which is not on the X chromosome. The allele for the development of normal cones (pigment cells in the retinal layer of the eye), **B**, is dominant to the allele for no cone development, **b**.

The figure below shows the phenotypes of members of a different family in which both types of colour blindness occur.



Which of the following are possible genotypes for individuals **P**, **Q** and **S**?

	P	Q	S
A	BBX^GX^G	bbX^GY	BBX^gY
B	BbX^GX^g	BbX^gY	BbX^GY
C	BBX^gX^g	bbX^GY	BbX^gY
D	BbX^GX^g	BbX^gY	BBX^GY

19. In the magpie moth *Abraxas sp.*, the female is the heterogametic sex and the gene for wing colour is sex-linked.

In a cross between a normal coloured male and a pale coloured female, the F1 offspring consisted of all normal coloured individuals with the two sexes in equal proportions.

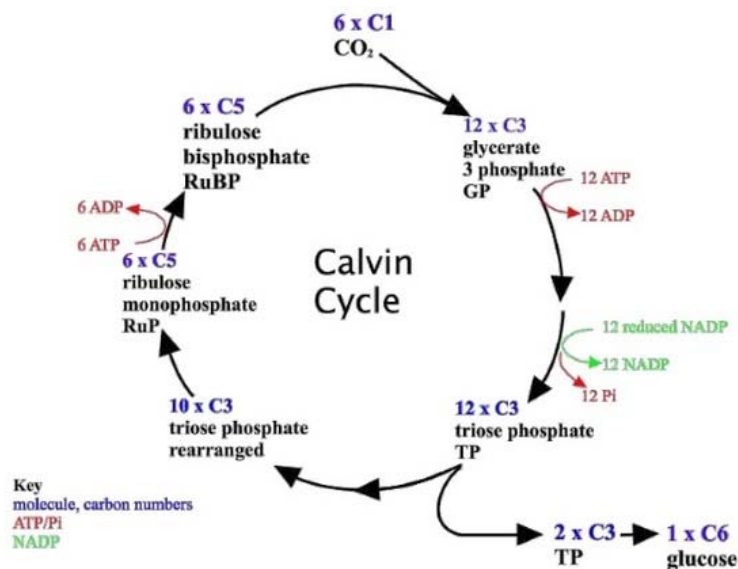
Which ratio would be obtained in the F2 generation produced from the F1 generation?

- A** Normal coloured males to normal females 1:1
- B** Normal coloured males and females to pale females 3:1
- C** Normal coloured males and females to pale males and females 1:1
- D** Normal coloured males to pale coloured females 1:1

20. Which of the following is the **most significant** reason for the effectiveness of nucleic acid hybridisation in highlighting a particular band of DNA on a gel electrophoresis slab?

- A The use of sponge and filter paper to draw buffer solution through the nitrocellulose paper containing the separated DNA
- B The use of radioactively labeled DNA primer sequences that specifically bind to the targeted sequences of DNA
- C The use of ethidium bromide that produces a strong reddish-yellow fluorescence when bands are exposed to UV radiation
- D The use of single stranded radioactively labelled probes complementary to the targeted bands

21. The following is a diagram of the Calvin cycle.



Assuming each turn starts with 1 CO₂ and 1 RuBP, how many turns of the Calvin cycle would be required to produce 40 molecules of glucose?

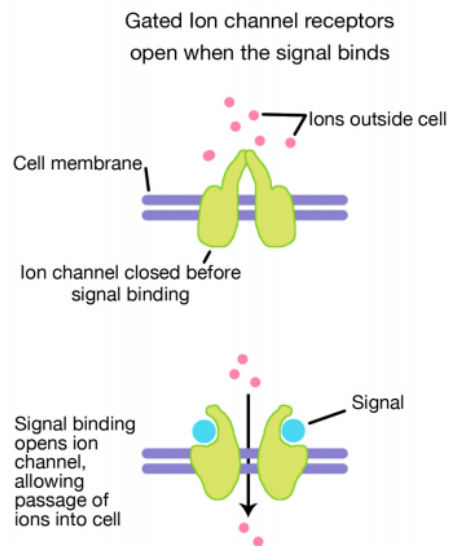
- A 40
- B 120
- C 240
- D 480

22. In living cells, 2,4 dinitrophenol acts as a proton ionophore, an agent that can shuttle protons across biological membranes.

Which of the following is **not** a possible consequence of the introduction of 2,4 dinitrophenol into an animal cell?

- A Less oxygen is taken up by the cell.
- B The proton gradient across the inner mitochondrial membrane is dissipated.
- C The rate of glycolysis in the cell will increase.
- D The rate of Krebs cycle in the cell will increase.

23. The following is a diagram of an ion channel receptor and how it works.



This is a type of cell signaling receptor found in nerve cells. When a signal molecule binds to these receptors, the receptor changes conformation and causes the influx of ions into nerve cells. This directly causes the formation of an electrical action potential, which is transmitted along the nerve cell.

Which of the following statements **incorrectly** compares ion channel receptors with a G protein linked receptor?

- A Both ion channel and G protein linked receptors involve the activation of genes.
- B Both ion channel and G protein linked receptors involve the change of conformation of a receptor protein.
- C Ion channel receptors do not involve a second messenger.
- D G protein linked receptors do not allow the passage of ions into the cell.

24. New research conducted by evolutionary biologists worldwide paints cities as evolutionary "change agents", says a trio of biologists from the University of Toronto Mississauga (UTM).

A compilation of 15 new research papers, published as a special issue of *Proceedings of the Royal Society B: Biological Sciences*, confirms that cities frequently alter evolution by natural selection.

The following statements are possible ways in which cities could alter evolution by natural selection.

- 1 Cities are generally warmer than natural areas and thus organisms adapted to higher temperatures would be selected for within cities.
- 2 Cities release large amounts of environmental pollutants and thus organisms with greater resistance to common pollutants would be selected for in areas within cities.
- 3 Cities provides additional food sources for many organisms and thus organisms that are adapted to feed on a wider range of food types would be selected for within cities.
- 4 Cities have a large human population so organisms that are better able to interact with humans will be selected for within the cities.

Which statements are potentially correct?

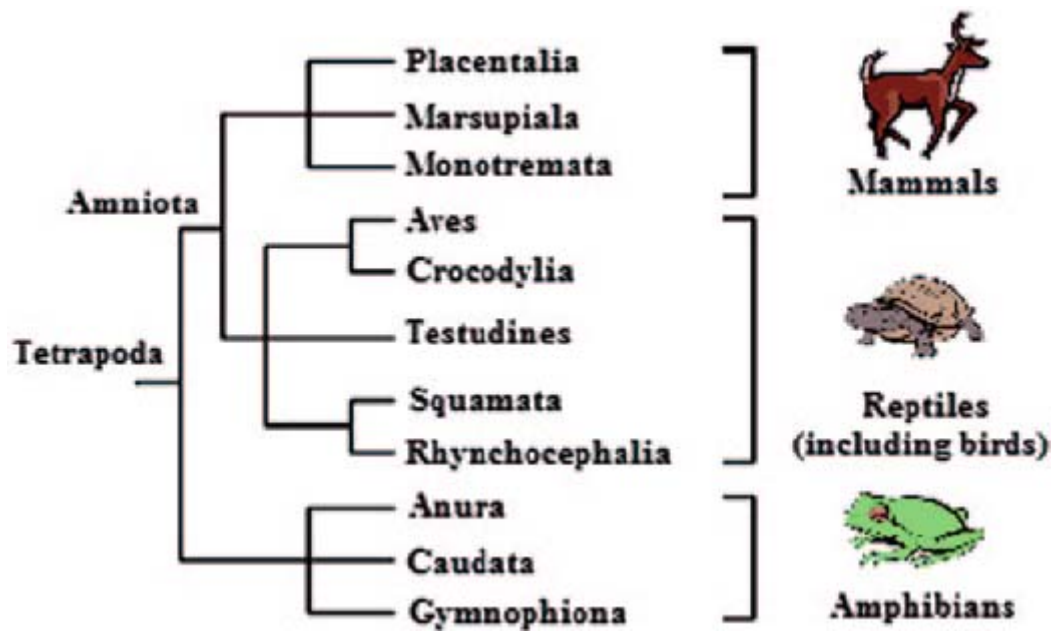
- A 1 and 2 only
- B 2 and 4 only
- C 1, 2 and 3 only
- D 1, 2, 3 and 4

25. Lizards in the secondary forests of Pulau Ubin have always been classified as the same species as the lizards on mainland Singapore. However, it is suspected that the lizards on Pulau Ubin have recently evolved into a new species.

Which of the following scientific investigations is **least likely** to confirm this suspicion?

- A A detailed comparison of lizard fossils obtained from both the Pulau Ubin secondary forests and the mainland.
- B Aligning and studying a homologous region of mitochondrial DNA obtained from both Pulau Ubin and mainland lizards.
- C A study of the habitats occupied by lizards in both locations and their interactions with both the biotic and abiotic components of their environment.
- D An in-depth comparative study of the primary morphological and physiological features of lizards from both locations.

26. The following figure shows the phylogenetic tree of some four-legged organisms.



Based only on the information in the phylogenetic tree, which of the following statements is **incorrect**?

- A** Mammals are more closely related to reptiles than they are to amphibians.
- B** There has been a greater amount of time for tetrapods to evolve compared with the group Amniota.
- C** When comparing homologous DNA regions from an organism in the group Squamata and Rhynchocephalia, there will be fewer differences than when comparing Squamata with Crocodylia.
- D** Placental mammals and Marsupials are very similar but they evolved slightly differently due to their separation by geography.

27. The following table depicts some ways in which drugs can be used to target immune system cells.

Target	Principal immune function	Source(s)	Principal pathway mechanism(s)	Select drug(s) in clinical investigation
B7-H3	Inhibitory	Transmembrane receptor protein found on APCs, tumor cells, host cells	Binds to unknown receptors on T cells causing inhibitor immune signals; however, primarily inhibitory	Enoblituzumab (MGA271)
ICOS	Inhibitory	Transmembrane receptor protein found on T cells (highly expressed on T-regulatory cells)	Exerts an immune inhibitory function by binding to ICOS-L on APCs stimulating T-regulatory cell function, thus mediating overall immune suppression	MEDI570
OX40	Stimulatory	Transmembrane receptor protein found on T cells	Binds to OX40-L on APCs to stimulate T-cell proliferation and activity	MEDI6469 MOXR0916
GITR	Stimulatory	Transmembrane receptor found on T cells	Produces stimulatory signal upon binding to GRITL on APCs stimulating T-cell proliferation and activity; also involved in T-regulatory cell function	TRX518

Which drug(s) could possibly be used in the clinical treatment of cancerous T cells?

- A MEDI570 only
- B MEDI6469 only
- C MGA271 and TRX518 only
- D MEDI570 and MOXR0916 only

28. Over the past month, a team from the Ethiopian Wolf Conservation Programme has suggested the implementation of the first oral vaccination campaign to pre-empt outbreaks of rabies among Ethiopian wolves, the world's most endangered canid, in their stronghold in the Bale Mountains of southern Ethiopia.

Which of the following is the most biologically accurate prediction of the effect of this oral vaccination attempt?

- A The vaccination will not work as vaccines only work for humans.
- B The vaccination will not work as the wolves are likely to develop resistance against the vaccine.
- C The vaccination will work as it contains antibodies that will target the rabies pathogen.
- D The vaccination will work as wolves will produce plasma cells to target the rabies pathogen.

29. It has been discovered that deep corals, which are found at ocean depths below the reach of sunlight, is also affected by climate change.

Which of the following statements is a valid explanation as to why this is so?

- A The rate of photosynthesis at the deeper waters inhabited by the deep coral species is inhibited by a lack of carbon dioxide.
- B Deep water coral species are adapted to lower temperatures and are unable to migrate to shallow waters which have a higher water temperature.
- C The warming of surface water temperatures due to global warming has led to even the deeper waters heating up beyond the natural range of deep water coral species.
- D The rising sea levels globally have led to deep corals being unable to receive sunlight for use by its symbiotic algae.

30. All of the following are possible impacts of global warming except

- A heavier strain on global food supplies such as livestock.
- B increased transmission of diseases such as tuberculosis and HIV.
- C larger populations of crop pests in the tropical and sub-tropical regions.
- D smaller reservoir for biomedicines due to reduced biodiversity of the tropics.

END OF PAPER